



# Engine & Cooling Basics

## Brief Overview

This note covers Internal Combustion Engines and Refrigeration & Air Conditioning and was created from a 26-page PDF from the source URL. The notes provide a quick tour of engine types, cycle mechanics, performance metrics, refrigerant selections, and basic AC system design.

## Key Points

- Engine classifications and cycle diagrams
  - **Performance formulas** for mean effective pressure, brake power, and efficiencies
  - Refrigeration cycles and refrigerant properties
  - Air-conditioning system components and operation flow
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## Internal Combustion Engines

### Introduction

An **internal combustion engine** (I.C. Engine) is a heat engine that converts heat energy released by combustion of fuel inside the engine cylinder into mechanical energy.

**Definition:** I.C. Engine - A heat engine where fuel combustion occurs within the engine cylinder itself

### Advantages:

- High efficiency
- Light weight
- Compactness
- Easy starting
- Adaptability
- Suitable for locomotives

**Examples:** Petrol engines, Diesel engines, Gas engines

## Classification of I.C. Engines

### 1. Based on Fuel Type

- **Petrol engines:** Use petrol as fuel
- **Diesel engines:** Use diesel as fuel

### 2. Based on Number of Strokes

- **4-stroke engine:** Working cycle completed in 4 strokes
- **2-stroke engine:** Working cycle completed in 2 strokes

### 3. Based on Ignition Method

- **Spark ignition (S.I.):** Fuel ignited by electric spark
- **Compression ignition (C.I.):** Ignition by high compression

### 4. Based on Combustion Cycle

- **Otto cycle:** Combustion at constant volume
- **Diesel cycle:** Combustion at constant pressure

### 5. Based on Number of Cylinders

- **Single cylinder:** Only one cylinder
- **Multi-cylinder:** 2, 3, 4, 6, or 8 cylinders

### 6. Based on Cylinder Arrangement

- **Inline/Parallel:** Cylinders arranged in a line
- **Radial:** Cylinders arranged radially
- **V-engine:** Two cylinders at an angle
- **Opposed:** Cylinders arranged opposite

### 7. Based on Cooling Method

- **Air-cooled:** Cooling by air (bikes, scooters)
- **Water-cooled:** Cooling by water with radiator (cars)

## IC Engine Components

The following diagram illustrates major parts of an IC Engine:

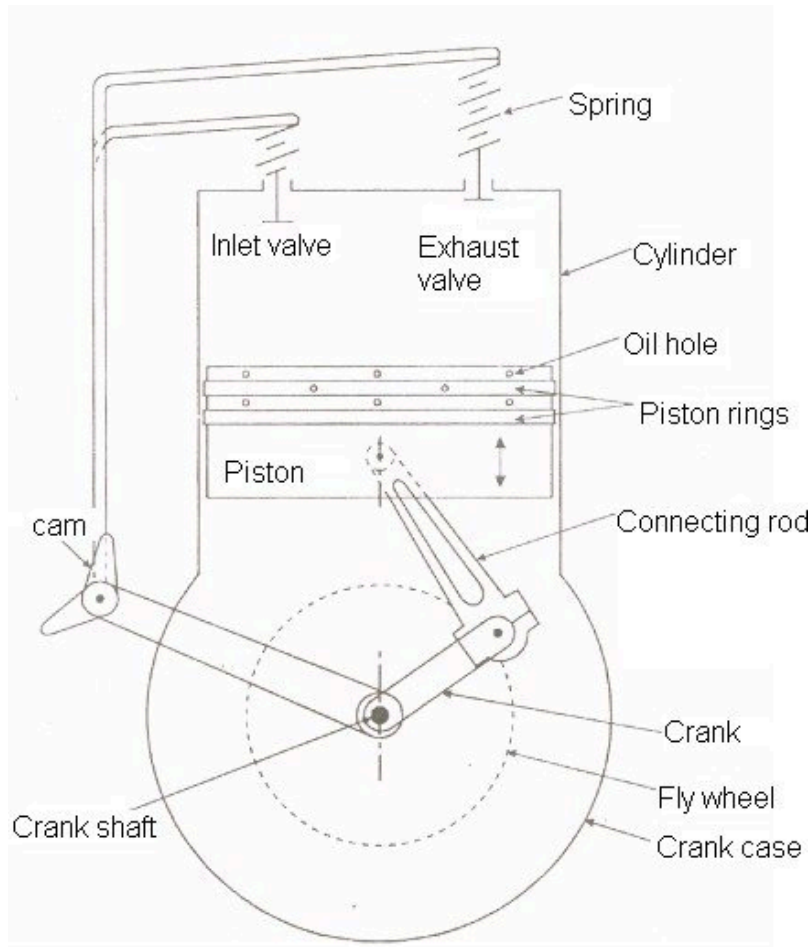


Fig: Parts of IC Engine

### Key Components and Functions:

#### Cylinder (Cylinder Block):

- Main part where combustion occurs
- Withstands high gas pressure up to  $2800^{\circ}\text{C}$
- Made of grey cast iron
- Cooled by fins (air) or water jackets

#### Cylinder Head:

- Fitted on top of cylinder block
- Contains inlet valve, exhaust valve, spark plug/fuel injector
- Gasket prevents gas leakage (copper/asbestos)

#### Piston:

- Cylindrical plug converting heat to mechanical energy
- Made of aluminum alloy

- Functions:
  - Acts as seal
  - Provides heat flow passage
  - Transmits force to crankshaft

### Connecting Rod:

- Links piston to crankshaft
- Converts reciprocating to rotary motion
- I-beam cross-section for rigidity

### Crankshaft:

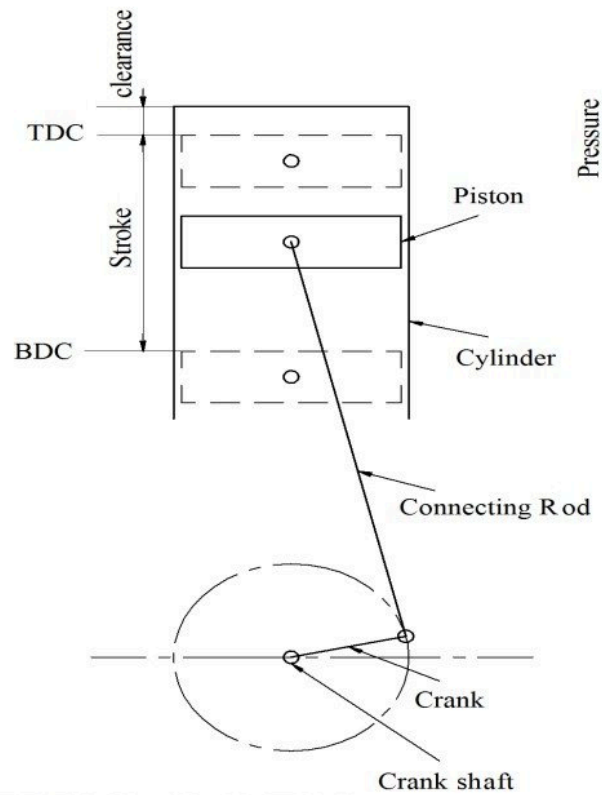
- Receives power from connecting rod
- Fixed in crankcase
- Connected to clutch

### Crankcase:

- Bottom of cylinder block
- 2-stroke: sealed and airtight
- 4-stroke: lubricating oil reservoir

## IC Engine Terminology

The following diagram shows piston-cylinder assembly with key terminology:



1. **Bore**: Nominal inside diameter of engine cylinder
2. **Stroke**: Linear distance between extreme piston positions
3. **Top Dead Centre (TDC)**: Extreme position near cylinder head (minimum volume)
4. **Bottom Dead Centre (BDC)**: Extreme position towards crank end (maximum volume)
5. **Compression ratio**:  $r = \frac{V_s + V_c}{V_c}$ 
  - Petrol engine: 7:1 to 12:1
  - Diesel engine: 16:1 to 22:1

## 4-Stroke Petrol Engine

Works on **Otto cycle** (constant volume combustion). Known as **Spark Ignition (S.I.)** engine.

### Four Strokes:

1. **Suction Stroke** (0°-180°):
  - Inlet valve open, exhaust closed
  - Piston TDC to BDC
  - Draws petrol-air mixture from carburetor

2. **Compression Stroke** (180°-360°):

- Both valves closed
- Piston BDC to TDC
- Mixture compressed, spark at end

3. **Power/Expansion Stroke** (0°-180°):

- Both valves closed
- Piston TDC to BDC
- Combustion forces piston, produces power

4. **Exhaust Stroke** (180°-360°):

- Exhaust valve open, inlet closed
- Piston BDC to TDC
- Burnt gases expelled

## 4-Stroke Diesel Engine

Works on **Diesel cycle** (constant pressure combustion). Known as **Compression Ignition (C.I.)** engine.

### Four Strokes:

1. **Suction Stroke**: Pure air drawn in
2. **Compression Stroke**: Air compressed, fuel injected at end (auto-ignition)
3. **Power Stroke**: Combustion forces piston
4. **Exhaust Stroke**: Burnt gases expelled

## 2-Stroke Petrol Engine

One cycle completed in 2 strokes (one crankshaft revolution). Uses ports instead of valves.

### Ports:

- **Inlet port**: Fresh charge entry
- **Transfer port**: Charge transfer to piston top
- **Exhaust port**: Hot gases exit

### Two Strokes:

1. **Upward Stroke**:

- Piston BDC to TDC
- Compression occurs
- Fresh charge drawn to crankcase

- Spark ignition at TDC

## 2. Downward Stroke:

- Piston TDC to BDC
- Power production
- Ports uncovered, gases escape
- Fresh charge enters cylinder

## Engine Comparisons

### Petrol vs Diesel Engine

| Aspect            | Petrol Engine    | Diesel Engine     |
|-------------------|------------------|-------------------|
| Ignition          | Spark plug       | Compression       |
| Cycle             | Otto             | Diesel            |
| Fuel              | Petrol           | Diesel            |
| Suction           | Air-fuel mixture | Only air          |
| Combustion        | Constant volume  | Constant pressure |
| Compression ratio | 7:1-12:1         | 16:1-22:1         |
| Fuel cost         | High             | Low               |
| Power output      | Less             | More              |

### 2-Stroke vs 4-Stroke Engine

| Aspect            | 2-Stroke         | 4-Stroke             |
|-------------------|------------------|----------------------|
| Cycle completion  | 2 strokes        | 4 strokes            |
| Power development | Every revolution | Alternate revolution |
| Flywheel          | Not essential    | Essential            |
| Valves/Ports      | Ports            | Valves               |
| Charge admission  | Via crankcase    | Direct to cylinder   |
| Construction      | Simple           | Complex              |

## Performance Parameters

### 1. Mean Effective Pressure (pm):

Average pressure during power stroke

- Units: N/m<sup>2</sup>

### 2. Indicated Power (IP):

- Power developed within cylinder
- $IP = \frac{P_m \times L \times A \times n}{60 \times 1000}$  kW
- Where:  $P_m$  = mean effective pressure,  $L$  = stroke length,  $A$  = piston area,  $n$  = cycles/min

### 3. Brake Power (BP):

- Power available at crankshaft
- $T = (W - S) \times R$
- $BP = \frac{2\pi NT}{60 \times 1000}$  kW

### 4. Frictional Power (FP):

- $FP = IP - BP$

### 5. Mechanical Efficiency:

- $\eta_{mech} = \frac{BP}{IP} \times 100$

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## Refrigeration and Air Conditioning ❄️

### Introduction to Refrigeration

**Definition:** Refrigeration is a method of reducing system temperature below surroundings and maintaining it by continuously abstracting heat

**Refrigerator:** Device producing refrigeration effect

### Concepts of Refrigeration

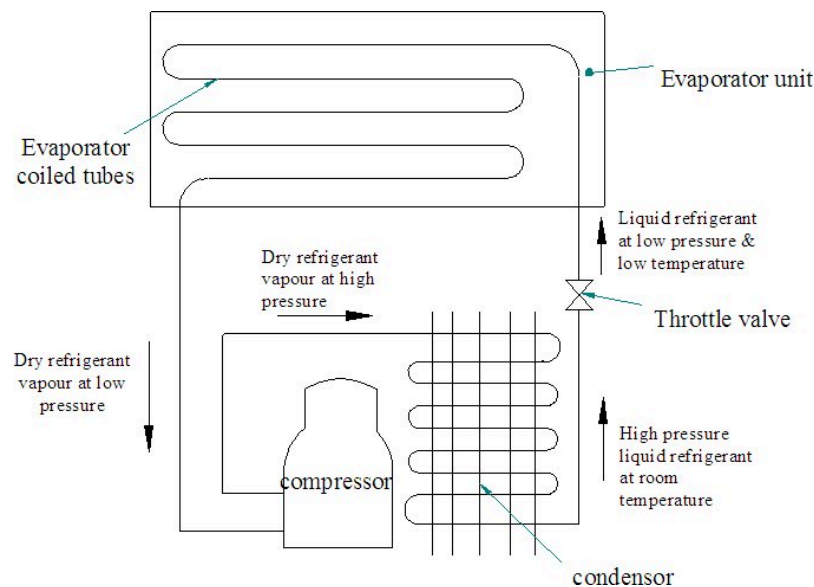
1. Heat transfers from high to low temperature
2. Heat transfer from low to high temperature requires external means
3. Working fluid changes phase: vapor  $\leftrightarrow$  liquid
4. Liquid to vapor phase change creates cooling effect

### Principle of Refrigeration

Removes heat from enclosed space by exploiting refrigerant's phase change and pressure-temperature relationship.

**Common Refrigerants:** Freon, Ammonia, CO<sub>2</sub>, SO<sub>2</sub>, hydrocarbons, methylene chloride, ethylene, ethane

## Refrigerator Components



### 1. **Compressor:**

- Compresses low T&P vapor to high T&P
- Power-absorbing device driven by motor

### 2. **Condenser:**

- U-tube coils
- Refrigerant rejects heat to surroundings
- Phase change: vapor → liquid

### 3. **Expansion Valve:**

- High P&T liquid expands to low P&T
- Temperature drops due to partial evaporation

### 4. **Evaporator:**

- U-tube cooling coils
- Low T mixture absorbs heat, changes to vapor

## Properties of Good Refrigerant

1. Very low boiling and freezing points
2. High enthalpy of evaporation, low specific volume

3. High latent heat of evaporation
4. Low saturation pressure
5. Good thermal conductivity
6. Non-toxic
7. Non-flammable
8. Non-corrosive
9. Economical
10. Chemically stable

### Common Refrigerants Table

| No. | Refrigerant                | Properties                                                                                                                        |
|-----|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| 1   | Ammonia (NH <sub>3</sub> ) | Boiling point: -33.3°C. Highly flammable, irritating, corrosive. Used in large industries, cold storage, ice plants               |
| 2   | CO <sub>2</sub>            | Non-toxic, non-flammable. Boiling point: -77.6°C. Compact plant size. Used in ships                                               |
| 3   | SO <sub>2</sub>            | Colorless, suffocating odor. Boiling point: -10°C. Used in old household refrigerators                                            |
| 4   | Freon-12                   | Non-flammable, non-explosive, non-corrosive, odourless. Boiling point: -29.8°C. Used in domestic refrigerators, water coolers, AC |
| 5   | Freon-22                   | Boiling point: -40.8°C. High-pressure refrigerant. Used in large AC, food freezing                                                |

### Vapour Compression Refrigerator

Working cycle:

1. Liquid refrigerant absorbs heat in evaporator → vapor
2. Low T&P vapor compressed → high T&P
3. High T&P vapor condenses in condenser → liquid
4. High P liquid expands in expansion valve → low P&T
5. Cycle repeats continuously

## Air Conditioning 🌬️

**Definition:** Process of simultaneous control of temperature, humidity, cleanliness in confined space

### Principle of Air Conditioning

Machine draws warm air, cools it by refrigeration, circulates cooled air continuously to maintain required indoor temperature.

### Applications

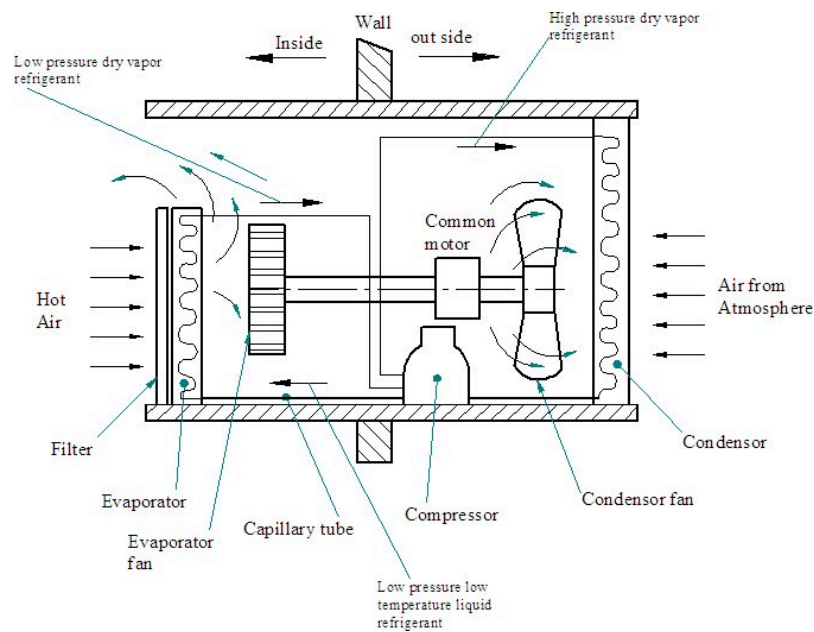
#### 1. Comfort Applications

- Residential: Houses, apartments
- Institutional: Offices, hospitals
- Commercial: Shopping centers, malls
- Transportation: Aircraft, automobiles, ships

#### 2. Process Applications

- Hospitals: Operation theatres
- Clean rooms: IC production, pharmaceuticals
- Laboratories: Animal breeding
- Industrial: Nuclear facilities, food processing, data centers

### Room Air Conditioner



### Components:

- Air filter
- Double shaft motor (drives fan and blower)
- Insulated partition separating room/outdoor sides

### Working of Room AC

1. **Blower** sucks warm room air through filter and evaporator
2. **Evaporator coils** cool air, refrigerant evaporates
3. **Compressor** draws and compresses refrigerant vapor
4. **Condenser fan** draws atmospheric air through condenser coils
5. **Condenser** cools refrigerant, changes to liquid
6. **Capillary tube** expands refrigerant
7. **Cycle repeats** continuously